

Semantic Obstruction and Irreducible Residue

A Category-Theoretic Framework for Representation-Invariant Constraints, Traversability, and Local-to-Global Failure

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Abstract

This paper introduces a general framework for analyzing constraints relative to representation. A semantic obstruction is defined as obstruction measured within a chosen representational scheme, while the irreducible residue is the orbit-minimized obstruction over all admissible re-descriptions. This yields a canonical decomposition of obstruction into representation-dependent and representation-invariant components, establishes orbit invariance, proves existence under mild compactness and lower-semicontinuity hypotheses, and produces a trichotomy classifying system-target pairs as removable, irreducible and fatal, or irreducible and traversable. The framework is formulated categorically in terms of a representation groupoid, then lifted to gauge-theoretic, sheaf-theoretic, and higher-categorical settings. Worked examples are given in gauge theory, nonholonomic control, sheaf gluing, and optimization. A final appendix shows how the framework can serve as a universal front-end for broad mathematical research programs, including but not limited to work on Yang-Mills, Navier-Stokes, the Riemann Hypothesis, P versus NP, and structurally similar problems. The framework is general and does not itself solve any particular open problem; rather, it formalizes a reusable invariant structure for distinguishing removable obstruction from irreducible but potentially traversable residue.

Keywords

semantic obstruction, irreducible residue, obstruction theory, representation groupoid, gauge invariance, cocycle obstruction, traversability, local-to-global failure, category theory, higher category theory, nonholonomic control, sheaf theory

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1. Introduction

Many hard problems arise through an interplay between structure and description. A system may appear blocked because it has been placed in a poor coordinate system, an inconvenient encoding, or an unsuitable local model. In that case, the obstruction is not intrinsic to the system alone; it is relative to the representation. On the other hand, some obstruction remains no matter how the system is re-described within the admissible class. That persistent remainder is the real structural content of the problem.

The aim of this paper is to formalize that distinction.

The central thesis is simple:

The mathematically meaningful part of an obstruction is not its value in any one representation, but the minimum obstruction that survives across all admissible re-descriptions.

This minimum is called the irreducible residue.

Once this is formalized, a new trichotomy becomes visible. A system-target pair may be:

1. removable, meaning the obstruction can be driven to zero by admissible re-description;
2. irreducible and fatal, meaning a positive residue remains and no successful admissible traversal exists;
3. irreducible and traversable, meaning a positive residue remains but the target can still be reached by an admissible representation-action pair.

The third regime is especially important. In practice, many systems are not made successful by removing all obstruction, but by aligning with what cannot be removed. This phenomenon appears in control, topology, gauge theory, engineering, and computation. The present framework is built to formalize precisely that regime.

The paper proceeds as follows. Section 2 defines the representation groupoid and obstruction functional. Section 3 introduces the irreducible residue and apparent obstruction. Section 4 proves the first structural results: decomposition, invariance, monotonicity, and existence. Section 5 introduces traversability and proves the trichotomy. Section 6 develops the categorical formulation. Section 7 specializes to cocycle and sheaf-theoretic obstruction. Section 8 gives a gauge-theoretic formulation. Section 9 lifts the theory to higher-categorical language. Section 10 gives worked examples with proofs. Section 11 discusses design principles and interpretation. Section 12 concludes. Appendix A provides an implementation guide for using the framework as a universal front-end to major research programs and analogous problem families.

2. Basic Setup

2.1. System states, targets, and representations

Let XX be a set, topological space, manifold, moduli space, or other admissible class of system states.

Let TT denote a target. The target may be: - a property to be achieved, - a simplification to be realized, - a traversal objective, - a consistency condition, - a desired construction, - a regularity condition, - a reduction in complexity, - or a spectral, topological, or dynamical requirement.

For each $x \in X$, let $\mathcal{R}(x)$ be a nonempty collection of representations of xx . A representation may be: - a coordinate chart, - a local trivialization, - an encoding, - a decomposition, - a model, - a computational description, - a gauge choice, - a coarse-graining, - or any other admissible mode of description.

We write $r \in \mathcal{R}(x)$ for a representation of xx .

2.2. The representation groupoid

We assume the admissible changes of representation are organized into a groupoid $\mathcal{G}$, called the representation groupoid.

For each fixed $x \in X$, the representations $\mathcal{R}(x)$ form a connected component, or orbit, in $\mathcal{G}$. Thus: - objects of $\mathcal{G}$ are representations, - morphisms are admissible re-descriptions, - every morphism is invertible.

We say $r_1 \sim r_2$ if r_1 and r_2 lie in the same orbit of $\mathcal{G}$.

The point of using a groupoid rather than a group is that different states may admit different representation classes, and not every representation needs to be globally comparable to every other one.

2.3. Obstruction functional

Fix a target TT .

An obstruction functional is a map $O: \{(r,x): x \in X, r \in \mathcal{R}(x)\} \rightarrow [0, \infty]$ $O: \{(r,x) : x \in X, r \in \mathcal{R}(x)\} \rightarrow [0, \infty]$ which we write as $O(r,x;T)$. $O(r,x;T)$.

Interpretationally, $O(r,x;T)$ measures how obstructed the target TT is when the system xx is viewed through the representation rr .

Depending on the domain, OO may measure: - failure of commutativity, - cocycle defect, - nontrivial holonomy, - curvature residue, - nontrivial topological class, - cost of control, - optimization infeasibility, - descriptive complexity, - symmetry mismatch, - concentration defect, - unresolved singularity signal, - lower-bound witness, - or another obstruction quantity.

No single interpretation is built into the definition.

3. Irreducible Residue and Apparent Obstruction

3.1. Irreducible residue

Definition 3.1

Let $x \in X$. The irreducible residue of xx relative to target TT is

$$\mathcal{I}(x;T) := \inf_{r \in \mathcal{R}(x)} O(r,x;T) \quad \boxed{\mathcal{I}(x;T) := \inf_{r \in \mathcal{R}(x)} O(r,x;T)}$$

This is the minimal obstruction remaining after optimizing over all admissible representations of xx .

If $\mathcal{I}(x;T) = 0$, then the obstruction is removable at the representational level.

If $\mathcal{I}(x;T) > 0$, then some positive obstruction survives every admissible re-description.

3.2. Apparent obstruction

Definition 3.2

For $r \in \mathcal{R}(x)$ such that $O(r, x; T) < \infty$, define the apparent obstruction by $\mathcal{A}(r, x; T) := O(r, x; T) - \mathcal{I}(x; T)$. $\mathcal{A}(r, x; T) := O(r, x; T) - \mathcal{I}(x; T)$.

This is the portion of the obstruction visible in representation r that is not forced by the orbit-minimized residue.

The decomposition $O(r, x; T) = \mathcal{A}(r, x; T) + \mathcal{I}(x; T)$ $O(r, x; T) = \mathcal{A}(r, x; T) + \mathcal{I}(x; T)$ is the basic algebraic split between representational artifact and structural remainder.

4. Main Structural Theorems

4.1. Canonical decomposition

Theorem 4.1

For every $x \in X$, target T , and representation $r \in \mathcal{R}(x)$ with finite obstruction, $O(r, x; T) = \mathcal{A}(r, x; T) + \mathcal{I}(x; T)$, $O(r, x; T) = \mathcal{A}(r, x; T) + \mathcal{I}(x; T)$, with $\mathcal{A}(r, x; T) \geq 0$. $\mathcal{A}(r, x; T) \geq 0$.

Proof

By definition of infimum, $\mathcal{I}(x; T) \leq O(r, x; T) - \mathcal{I}(x; T) \leq O(r, x; T)$ for every $r \in \mathcal{R}(x)$. Therefore $O(r, x; T) - \mathcal{I}(x; T) \geq 0$. $O(r, x; T) - \mathcal{I}(x; T) \geq 0$. By Definition 3.2, this difference is precisely $\mathcal{A}(r, x; T)$. Rearranging yields the decomposition. QED.

4.2. Orbit invariance

Theorem 4.2

The irreducible residue $\mathcal{I}(x; T)$ is invariant under admissible re-description. Equivalently, it depends only on the orbit $\mathcal{R}(x) \subseteq \mathcal{G}$.

Proof

The definition of $\mathcal{I}(x;T)$ takes the infimum over all representations in the orbit $\mathcal{R}(x)$. Any admissible transformation within that orbit preserves the set over which the infimum is taken. Therefore the value of the infimum is unchanged. QED.

4.3. Monotonicity under restriction of admissible class

Theorem 4.3

Let $\mathcal{R}_1(x)$ and $\mathcal{R}_2(x)$ be two admissible representation classes for the same state x , with $\mathcal{R}_2(x) \subseteq \mathcal{R}_1(x)$. Then $\mathcal{I}_{\mathcal{R}_2}(x;T) \geq \mathcal{I}_{\mathcal{R}_1}(x;T)$.

Proof

The infimum over a smaller set cannot be less than the infimum over a larger set. QED.

4.4. Existence of minimizing representation

Theorem 4.4

Assume: 1. $\mathcal{R}(x)$ is compact, 2. $r \mapsto O(r,x;T)$ is lower semicontinuous.

Then there exists $r^* \in \mathcal{R}(x)$ such that $O(r^*,x;T) = \mathcal{I}(x;T)$.

Proof

A lower semicontinuous function on a compact space attains its infimum. QED.

Remark

This theorem supplies a standard sufficient condition for the irreducible residue to be realized by an actual representation rather than only as a limiting value.

5. Traversability and the Trichotomy

5.1. Success predicate and admissible actions

Let $\mathcal{U}(x)$ be a set of admissible actions, controls, or procedures available for state x .

Let $P(r,u,x;T)$ be a success predicate taking values in $\{\text{true}, \text{false}\}$, or more generally a performance functional whose threshold determines whether the target has been achieved.

5.2. Traversability

Definition 5.1

We say that (x,T) is traversable if there exist $r \in \mathcal{R}(x), u \in \mathcal{U}(x)$ such that $P(r,u,x;T)$ holds.

The key point is that traversability is logically independent of whether $\mathcal{I}(x;T)$ vanishes.

5.3. Trichotomy

Theorem 5.2

For every system-target pair (x,T) , exactly one of the following holds:

1. Removable: $\mathcal{I}(x;T)=0$.
2. Irreducible and fatal: $\mathcal{I}(x;T)>0$ and (x,T) is not traversable.
3. Irreducible and traversable: $\mathcal{I}(x;T)>0$ and (x,T) is traversable.

Proof

Either $\mathcal{I}(x;T)=0$ or $\mathcal{I}(x;T)>0$. In the first case, we are in case (1). In the second case, either there exists an admissible successful pair (r,u) , or none exists. These alternatives are mutually exclusive and exhaustive, giving cases (2) and (3). QED.

Interpretation

This theorem formalizes the third regime that is frequently missing in standard treatments: non-removable obstruction need not imply impossibility.

6. Category-Theoretic Formulation

6.1. Obstruction as an order-valued functor

Let $([0, \infty], \leq)$ and $([0, \infty], \leq)$ be viewed as a thin category.

Then an obstruction assignment may be regarded as a functor-like structure $O_T: \mathcal{R}(x) \rightarrow ([0, \infty], \leq)$ $O_T: \mathcal{R}(x) \rightarrow ([0, \infty], \leq)$ provided one imposes a compatibility condition on morphisms. In the most basic form, the functor sends each object r to the value $O(r, x; T)$. If morphisms preserve or monotone obstruction values, then O_T behaves as a genuine functor into an ordered target category.

The irreducible residue is then the lower envelope of this order-valued assignment on the connected component $\mathcal{R}(x)$.

6.2. Groupoid viewpoint

The groupoid viewpoint is especially natural. One should think of: - objects as admissible descriptions, - arrows as changes of description, - orbit invariants as structural content.

Then $\mathcal{I}(x; T)$ is simply an orbit invariant obtained by optimizing a nonnegative defect over the orbit.

6.3. Derived interpretation

Many classical obstructions arise from failure of a lifting, gluing, splitting, or commutativity condition. In categorical language, these appear as: - failure of a diagram to commute, - nonexistence of a global section, - nontriviality of an extension class, - nonvanishing derived functor, - nontrivial monodromy, - or obstruction class in a cohomology group.

The present framework does not replace those structures. Rather, it adds a new layer: it asks what obstruction remains after minimizing over all admissible reformulations of the same underlying system.

7. Sheaf-Theoretic and Cocycle Obstruction

This section gives the core local-to-global theorem.

7.1. Setup

Suppose x admits local descriptions over a cover $\mathcal{U} = \{U_i\}$, with transition data g_{ij} defined on overlaps $U_i \cap U_j$. Define the triple-overlap defect $\kappa_{ijk} := g_{ij} g_{jk} g_{ki}$.

In a consistent cocycle, $\kappa_{ijk} = e$ for all triple overlaps. Failure of this identity records a local-to-global obstruction.

7.2. Abstract cocycle obstruction theorem

Theorem 7.1

Assume: 1. The target T is global trivialization, flattening, or globally consistent gluing. 2. There exists a nonnegative functional f on cocycle defects such that $O(r, x; T) \geq f(\kappa_{ijk})$ for every admissible representation r . 3. $f(\kappa) = 0$ if and only if the cocycle defect vanishes in the admissible class.

If the cocycle class is nontrivial, then $\mathcal{I}(x; T) > 0$.

Proof

Suppose for contradiction that $\mathcal{I}(x; T) = 0$. Then for every $\varepsilon > 0$, there exists $r_\varepsilon \in \mathcal{R}(x)$ such that $O(r_\varepsilon, x; T) < \varepsilon$. By the lower bound hypothesis, $f(\kappa_{ijk}(r_\varepsilon)) \leq O(r_\varepsilon, x; T) < \varepsilon$. Hence $f(\kappa_{ijk}(r_\varepsilon)) \rightarrow 0$. By hypothesis (3), the cocycle defect must vanish in the admissible limit, contradicting nontriviality of the cocycle class. Therefore $\mathcal{I}(x; T) > 0$. QED.

Remark

This theorem captures the common pattern that failure of gluing produces a positive irreducible residue.

8. Gauge-Theoretic Formulation

The gauge-theoretic interpretation is a particularly important special case.

8.1. Local gauge data

Let: - local trivializations be the representations, - gauge transformations be admissible re-descriptions, - transition functions encode compatibility, - loop products or cocycles encode global obstruction.

Then semantic obstruction becomes a gauge-relative obstruction, and irreducible residue becomes the orbit-minimized gauge-invariant remainder.

8.2. Gauge-invariant residue theorem

Theorem 8.1

Assume the obstruction functional depends only on conjugacy-invariant data, such as: - holonomy around loops, - Wilson loop residue, - cocycle defect class, - curvature norms invariant under gauge.

Then the irreducible residue $\mathcal{J}(x;T)\backslash\mathcal{I}(x;T)$ is gauge-invariant.

Proof

Gauge transformations act by conjugation on loop products and cocycle representatives, but conjugacy-invariant defect functionals do not change under conjugation. Since the residue is the infimum over the whole gauge orbit, its value is unchanged by gauge transformation. QED.

8.3. Interpretation

This theorem captures the abstract skeleton of holonomy-based obstruction. The same pattern later specializes to normalized invariants in particular field theories, but the general structure already exists here.

9. Higher-Categorical Lift

The framework admits a natural lift to higher-categorical language.

9.1. Representation infinity-groupoid

Replace the representation groupoid $\mathcal{G}\backslash\mathcal{G}$ by an ∞ -groupoid $\mathcal{G}_\infty\backslash\mathcal{G}_\infty$, where: - 0-morphisms are representations, - 1-morphisms are admissible re-descriptions, - 2-morphisms are homotopies between re-descriptions, - higher morphisms encode coherence data.

The orbit structure is then understood homotopy-invariantly.

9.2. Higher obstruction assignments

An obstruction assignment may now take values in an ordered homotopical object $\mathcal{V}$, or be extracted from a spectral invariant, derived functor, or obstruction tower.

At the simplest level, one may still define $\mathcal{I}(x;T) := \inf_{r \in \pi_0(\mathcal{G}^\infty(x))} O(r,x;T)$, $\mathcal{I}(x;T) := \inf_{r \in \pi_0(\mathcal{G}^\infty(x))} O(r,x;T)$, where π_0 denotes connected components in the ∞ -groupoid.

9.3. Interpretation

The higher-categorical lift is useful when: - admissible transformations themselves have nontrivial homotopy, - coherence conditions matter, - obstruction arises through towers of extensions or lifting problems, - or one wants to compare obstructions up to higher equivalence rather than strict equality.

The basic residue remains an orbit-minimized obstruction, but now the orbit is homotopical rather than merely groupoidal.

10. Worked Examples with Full Proofs

This section gives concrete worked examples illustrating the framework.

10.1. Example 1: Gauge holonomy on a circle bundle

Let $M = S^1$. Consider a principal $U(1)$ -bundle with connection form A . Let the target T be global flattening, meaning zero holonomy around the base circle.

Representations are gauge choices $A \mapsto A + d\phi$. Let the obstruction be $O(A) := \inf_{n \in \mathbb{Z}} |\int_{S^1} A - 2\pi n|$. $O(A) := \inf_{n \in \mathbb{Z}} |\int_{S^1} A - 2\pi n|$.

This is the distance of the total holonomy phase from the nearest trivial $2\pi\mathbb{Z}$ -phase.

Claim

$O(A)$ is gauge-invariant and equals the irreducible residue.

Proof

Under gauge transformation, $A \mapsto A + d\phi$. Hence $\int S_1(A + d\phi) = \int S_1 A + \int S_1 d\phi = \int S_1 A$, $\int_{S^1} (A + d\phi) = \int_{S^1} A + \int_{S^1} d\phi = \int_{S^1} A$, since the integral of an exact form around a closed loop is zero. Therefore $O(A)$ is gauge-invariant. Since every gauge-equivalent representation has the same obstruction value, the infimum over the orbit equals that common value. Thus $\mathcal{I}(A; T) = O(A)$. If $\int S_1 A \notin 2\pi \mathbb{Z}$, then $\mathcal{I}(A; T) > 0$. QED.

Interpretation

This is the simplest instance of a semantic obstruction whose residue is already fully representation-invariant.

10.2. Example 2: Sheaf gluing on a two-set cover

Let $X = U \cup V$, with nonempty overlap $U \cap V$. Suppose one has local data s_U on U and s_V on V , and the target T is existence of a global section gluing the local pieces.

Let the obstruction be $O(r) := \|s_U|_{U \cap V} - s_V|_{U \cap V}\|$, $O(r) := \|s_U|_{U \cap V} - s_V|_{U \cap V}\|$, for some norm on the overlap data.

Claim

If the local data differ by a non-removable mismatch on the overlap, then $\mathcal{I}(x; T) > 0$.

Proof

If every admissible re-description preserves the overlap mismatch norm, then the obstruction functional is constant over the orbit. Hence the infimum equals that value. If that value is nonzero, the residue is positive. More generally, if re-description can reduce but not eliminate the mismatch, then the infimum remains positive. QED.

Interpretation

This example captures the basic logic of gluing defect without invoking advanced sheaf language.

10.3. Example 3: Nonholonomic control

Consider the standard nonholonomic system in $\mathbb{R}^3$ with coordinates (x, y, θ) , controlled by $\dot{x} = u_1 \cos \theta, \dot{y} = u_1 \sin \theta, \dot{\theta} = u_2$.
 $\dot{x} = u_1 \cos \theta, \quad \dot{y} = u_1 \sin \theta, \quad \dot{\theta} = u_2$.

The system cannot move directly in arbitrary directions at each instant. Let the target T be arbitrary local reachability.

Define the obstruction $O(r, x; T)$ to be the instantaneous directional deficiency in representation r . In any coordinate representation preserving the control structure, the rank deficiency is nonzero at the level of immediate motion. Hence the residue is positive: $\mathcal{I}(x; T) > 0$.

Yet the Lie bracket $[X_1, X_2]$ generates the missing direction, and the system is locally controllable.

Claim

This system is irreducible and traversable.

Proof

The instantaneous rank deficiency cannot be removed by admissible coordinate change preserving the control structure, so $\mathcal{I}(x; T) > 0$. However, Chow-Rashevskii theory implies local reachability from the bracket-generating property. Thus successful traversal exists despite positive residue. By Theorem 5.2, the system lies in the irreducible and traversable regime. QED.

Interpretation

This is a canonical worked example of irreducible obstruction without fatality.

10.4. Example 4: Optimization and convexification limit

Let $f: \mathbb{R}^n \rightarrow \mathbb{R}$ be a nonconvex objective, and let T be global convex solvability.

Suppose admissible representations are diffeomorphic reparameterizations Φ , so that in representation Φ , the objective becomes $f_\Phi := f \circ \Phi^{-1}$.

Define $O(\Phi) :=$ a nonconvexity measure of f_Φ , $O(\Phi) := \text{a nonconvexity measure of } f_\Phi$, for example the size of the negative part of the Hessian where defined.

Then $\mathcal{J}(f;T) := \inf_{\Phi} O(\Phi) \setminus \mathcal{I}(f;T) := \inf_{\Phi} O(\Phi)$ measures the minimal nonconvexity surviving admissible reparameterization.

Claim

If no admissible reparameterization convexifies the objective, then $\mathcal{J}(f;T) > 0 \setminus \mathcal{I}(f;T) > 0$.

Proof

If $\mathcal{J}(f;T) = 0 \setminus \mathcal{I}(f;T) = 0$, then by definition one could find admissible reparameterizations along which the nonconvexity measure tends to zero. Under any closure or attainment hypotheses sufficient for the problem, that would produce an effectively convex representation, contradicting the assumption. Therefore the residue is positive. QED.

Interpretation

This example shows how the framework applies even outside topology and gauge theory.

11. Relation to Prior Frameworks

This section situates the present work relative to well-known literatures.

11.1. Classical obstruction theory

Classical obstruction theory studies whether one may extend or section a partial construction, with obstructions often living in cohomology groups. The present framework does not replace obstruction theory. Instead, it adds a representational layer: one asks not only whether an obstruction exists, but whether it can be diminished or removed by admissible reformulation. The irreducible residue is the orbit-minimized shadow of classical obstruction.

11.2. Sheaf theory and descent

The gluing language in Section 7 aligns naturally with sheaf-theoretic descent. Čech cocycles, nontrivial cohomology classes, and descent obstructions are concrete sources of positive residue when the target is global consistency or trivialization.

11.3. Gauge theory

Gauge theory already organizes local descriptions into orbits under gauge equivalence. Holonomy, curvature, cocycle data, and topological charge supply natural obstruction functionals. The current framework extracts the general principle that the meaningful obstruction is the orbit-minimized gauge-invariant residue.

11.4. Control theory

Nonholonomic systems provide a paradigm case in which positive residue does not imply impossibility. The irreducible and traversable regime is standard in control practice but seldom elevated to a general structural principle.

11.5. Category theory and higher structures

The representation groupoid and its higher-categorical lift place the framework in a natural categorical setting. The irreducible residue can be understood as an orbit-invariant lower envelope across a category of admissible descriptions.

12. Design Principle and Interpretation

The framework suggests the following general principle.

Principle of Semantic Irreducibility

The meaningful obstruction in a problem is not $O(r,x;T)$ for one chosen representation rr , but the orbit-minimized quantity $\mathcal{I}(x;T)$.

Operational reformulation

If $\mathcal{I}(x;T)=0$, the obstruction is removable and one should search for a better representation.

If $\mathcal{I}(x;T)>0$, then the question changes. One must determine whether the problem is fatal or traversable.

This turns many design questions from: Can the obstruction be removed?
into: Does the orbit-minimized residue vanish, and if not, can traversal be organized around it?

13. Conclusion

We have introduced a general framework for semantic obstruction and irreducible residue. The framework formalizes: - representation-relative obstruction, - orbit-minimized structural remainder, - canonical decomposition, - orbit invariance, - existence under mild hypotheses, - local-to-global obstruction via cocycle defect, - and a trichotomy separating removable, fatal, and traversable regimes.

The framework is intentionally general. It does not solve any domain-specific open problem by itself. Instead, it provides a reusable front-end for formulating such problems in a way that cleanly separates representational artifact from structural content. Its central contribution is the rigorous formalization of the irreducible but traversable regime.

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Appendix A. Universal Front-End for Clay Problems and Related Programs

This appendix explains how the present framework can be used as a universal front-end for broad research programs without claiming that the abstract framework itself solves any one of them.

A.1. General front-end recipe

To apply the framework to a problem family, specify five ingredients:

1. System space XX
2. Target TT
3. Representation class $\mathcal{R}(x)$ and admissible groupoid $\mathcal{G}$
4. Obstruction functional OO
5. Success predicate PP

Once these are fixed, compute or estimate: $\mathcal{I}(x;T) = \inf_{r \in \mathcal{R}(x)} O(r,x;T)$.
 $\mathcal{I}(x;T) = \inf_{r \in \mathcal{R}(x)} O(r,x;T)$.

Then ask:

1. Does $\mathcal{I}(x;T) = 0$?
2. If not, is the pair fatal or traversable?
3. Under scale change or coarse-graining, does the residue persist, dilute, or amplify?

This gives a disciplined front-end before entering domain-specific technical machinery.

A.2. Yang-Mills-type program

A front-end formulation may be organized as follows.

System: local gauge data, holonomy data, or quantum field configurations.

Target: global flattening, mass-gap-compatible spectral control, or elimination of trivialization failure.

Representations: local trivializations, gauge choices, coarse-grained encodings, representation-theoretic descriptions.

Obstruction: loop holonomy defect, cocycle defect, Wilson loop residue, normalized character defect, or mean-zero contraction defect.

Residue: orbit-minimized nontrivial holonomy or normalized obstruction.

Front-end question: Does a nonzero normalized residue persist under admissible scale passage?

This does not prove the mass gap. What it does is stabilize the conceptual target and prevent confusion between raw small-scale defect and normalized structural content.

A.3. Navier-Stokes-type program

System: velocity fields, vorticity fields, or scale-localized fluid states.

Target: global regularity, suppression of blow-up, or elimination of unresolved singular concentration.

Representations: coordinate systems, scale decompositions, wavelet frames, geometric flow-adapted descriptions.

Obstruction: concentration defect, vortex stretching imbalance, non-removable cascade concentration, or singularity indicator.

Residue: minimal concentration obstruction surviving admissible reformulation.

Front-end question: Are observed blow-up signals representational artifacts, or is there an orbit-invariant concentration residue?

This does not settle regularity. It clarifies what kind of object one would need to show vanishes, remains bounded, or becomes traversable under a suitable reformulation.

A.4. Riemann-Hypothesis-type program

System: explicit formula data, zero distributions, spectral traces, phase-encoded arithmetic objects.

Target: critical symmetry, elimination of off-line spectral defect, or restoration of a canonical balanced description.

Representations: spectral formulations, transformed explicit formulas, geometric encodings, dynamical systems analogues, phase-unwrapped descriptions.

Obstruction: critical-line deviation functional, asymmetry residue, phase mismatch, or non-removable spectral imbalance.

Residue: the minimal spectral asymmetry surviving admissible reformulation.

Front-end question: Does the orbit-minimized spectral defect vanish?

This does not prove the hypothesis. It provides a formal way to distinguish representational asymmetry from intrinsic asymmetry.

A.5. P versus NP-type program

System: problem families, proof systems, encoding models, circuit classes.

Target: tractable solution class, equivalence collapse, or elimination of computational hardness witness.

Representations: different encodings, reductions, proof frameworks, circuit bases, algebraic lifts.

Obstruction: lower-bound witness, barrier phenomenon, nonuniform hardness defect, compression obstruction.

Residue: the minimal hardness obstruction surviving admissible representation change.

Front-end question: Are currently observed lower-bound barriers merely framework-dependent, or do they contain a positive irreducible residue across admissible encodings?

Again, this does not solve the problem. It gives a clean language for distinguishing representational barriers from invariant hardness.

A.6. General lessons from the front-end viewpoint

Across all these programs, the same workflow emerges.

Step 1: Define the target before choosing the representation.

Step 2: Describe the admissible representation orbit.

Step 3: Choose an obstruction functional that reflects the actual difficulty, not just a convenient proxy.

Step 4: Compute or estimate the irreducible residue.

Step 5: Decide whether the regime is removable, fatal, or traversable.

Step 6: Only then apply specialized analytic, algebraic, topological, or computational machinery.

This helps prevent the common failure mode in which one chases an obstruction that is merely representational while ignoring the orbit-invariant remainder.

A.7. Why this front-end is useful

The front-end is useful because it does three things at once.

First, it forces one to say explicitly what counts as an admissible reformulation.

Second, it separates the target from the current descriptive language.

Third, it converts vague claims like “this seems fundamental” or “this is probably an artifact” into a well-defined mathematical question about whether $\mathcal{I}(x;T)$ vanishes.

A.8. Meta-principle

The universal lesson is:

The right question is not whether an obstruction appears in one formulation, but whether its orbit-minimized residue survives all admissible formulations.

That sentence can function as the conceptual front-end to broad classes of difficult problems.

Appendix B. Submission Polishing Notes

B.1. Suggested titles

Here are several title variants, depending on tone.

Formal: Semantic Obstruction and Irreducible Residue: A Category-Theoretic Framework for Representation-Invariant Constraints

More conceptual: Irreducible Residue: A General Theory of Representation-Invariant Obstruction and Traversability

More applied: Semantic Obstruction, Traversability, and Local-to-Global Failure: A Unifying Framework

B.2. Suggested short title

Semantic Obstruction and Irreducible Residue

B.3. Suggested opening footnote

This paper develops a general formal framework. It does not claim to solve any particular open problem by abstract argument alone; rather, it isolates a reusable invariant structure intended to organize domain-specific proof programs.

B.4. Suggested arXiv categories

Primary candidates: - math.CT - math.AT - math-ph

Possible secondary cross-listing depending on final examples: - math.OC - cs.CC

B.5. Suggested author note

If you want a philosophical but disciplined positioning sentence:

This work is foundational rather than problem-specific. Its aim is to formalize a general mathematical structure underlying a wide range of local-to-global, gauge-relative, and representation-sensitive obstructions.

Appendix C. Copy-Paste Preservation Notes

To preserve the mathematics when copying into plain text or Markdown environments:

1. Keep all formulas either inline as $\dots$ or display as
$$\dots$$
.
2. Avoid nonstandard theorem environments if moving across editors.
3. Keep equation punctuation outside the LaTeX delimiters when possible.
4. Use named definitions in plain text headers rather than custom macros unless you control the target renderer.

Example safe style:

Definition 3.1. The irreducible residue is $\mathcal{I}(x;T) := \inf_{r \in \mathcal{R}(x)} O(r,x;T)$.
 $(x;T) := \inf_{r \in \mathcal{R}(x)} O(r,x;T)$.

This style will survive most copy-paste pipelines.